CI	SI	2-sided CV
90%	0.1	1.645
95%	0.05	1.96
99%	0.01	2.576

1. $\bar{X} = 52$, $S = 10$, $n = 100$

$H_0 : \mu = 50$ v $H_1 : \mu \neq 50$

$SE = \frac{S}{\sqrt{n}} = \frac{10}{\sqrt{100}} = \frac{10}{10} = 1 \Rightarrow SE(\bar{x})$

$Z\text{-stat.} = \frac{\bar{x} - \mu_0}{SE} = \frac{52 - 50}{1} = 2$

Compare $|Z|$ w/ each critical value:

α	CV	$\lvert Z\rvert > CV$?	$\therefore$
.1	1.645	Y	Reject H_0
.05	1.96	Y	Reject H_0
.01	2.576	N	Fail to Reject H_0

95% CI

$52 \pm 1.96 \times 1 =$

$(50.04, 53.95)$

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2. $\hat{S} = 3.1$ $\quad \hat{SE}(\hat{S}) = 1.3$ $\quad H_0 : S = 0$?

@ 90% $= 3.1 \pm 1.645(1.3) = (0.9615, 5.2385)$ → Reject ($\alpha = .1$)
0 not in Range

@ 95% $= 3.1 \pm 1.96(1.3) = (0.5520, 5.6480)$ → Reject ($\alpha = .05$)
0 not in Range

@ 99% $= 3.1 \pm 2.576(1.3) = (-0.2488, 6.4488)$ → fail to Reject ($\alpha = .01$)
0 in Range

Check: $Z = \frac{3.1 - 0}{1.3} = 2.385 > 1.645$ & 1.96 but < 2.576 ✓

3. $\bar{X} = 11.5$, $S = 3$, $n = 36$

$H_0 : \mu = 12$ v $H_1 : \mu \neq 12$

$$SE = \frac{3}{\sqrt{36}} = \frac{3}{6} = 0.5$$

95% CI $= 11.5 \pm 1.96(0.5) =$ (10.52, 12.48) → Fail to Reject ($\alpha = .05$)

12 inside Range

Check:

$$z: \frac{11.5 - 12}{0.5} = -1 \qquad |-1| < 1.96 \checkmark$$

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4. $\bar{X}_A = 25$, $S_A = 8$, $n_A = 100$

$\bar{X}_B = 22$, $S_B = 7$, $n_B = 100$

★ Independent Samples

$$\bar{X}_A - \bar{X}_B = 25 - 22 = 3$$

$$\hat{SE} = \sqrt{\frac{S_A^2}{n_A} + \frac{S_B^2}{n_B}} = \sqrt{\frac{64}{100} + \frac{49}{100}} = \sqrt{0.64 + 0.49} \approx 1.0630$$

@ 95% CI $= 3 + 1.96(1.0630) = (0.9165, 5.0835)$

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5. $\bar{X}_A = 14$, $S_A = 6$, $n_A = 81$

$\bar{X}_B = 12$, $S_B = 5$, $n_B = 64$

★ Independent Samples

$H_0 : \mu_A - \mu_B = 0$?

$$\bar{X}_A - \bar{X}_B = 14 - 12 = 2$$

$$SE = \sqrt{\frac{6^2}{81} + \frac{5^2}{64}} = \sqrt{\frac{36}{81} + \frac{25}{64}} = \sqrt{0.4444 + 0.3906} \approx 0.9138$$

$$z = \frac{2}{0.9138} \approx 2.189$$

$\alpha = .1$: $|z| > 1.645$ → Reject

$\alpha = .05$: $|z| > 1.96$ → Reject

$\alpha = .01$: $|z| < 2.576$ → Fail to Reject

6. $\bar{x}_A = 3.2$, $s_A = 1.5$, $n_A = 49$ ★ Independent Samples

$\bar{x}_B = 2.8$, $s_B = 1.4$, $n_B = 64$

$H_0: \mu_A - \mu_B = 0$

$\bar{x}_A - \bar{x}_B = 3.2 - 2.8 = 0.4$

$$SE = \sqrt{\frac{1.5^2}{49} + \frac{1.42^2}{64}} = \sqrt{\frac{2.25}{49} + \frac{1.96}{64}} = \sqrt{0.04592 + 0.03063} \approx 0.2767$$

$$z = \frac{0.4}{0.2767} \approx 1.446$$

@ 95% CI $0.4 \pm 1.96\,(0.2767) = (-0.1423, 0.9423)$

Since $|z| < 1.96$ & CI includes 0 we fail to Reject H_0 @ $\alpha = .05$. The observed gap of 0.4 is only about 1.4 standards errors from 0. The data doesn't show a statistically detectable difference in defect scores between the factories.